Vectors Quiz, Version B

NAME DATE PERIOD SCORE / 16

Show your work in the boxes and box your final answers with units. Round calculations to two significant figures. Angles on the 0–360° scale start at east and run counterclockwise: north is 90°, west is 180°, south is 270°. A negative angle runs clockwise from east. Any correct way of writing a direction counts.

Equations you may use

$R_x = R\cos\theta$ $R_y = R\sin\theta$ $\tan\theta = R_y / R_x$

$R^2 = R_x^2 + R_y^2$ $R_x = A_x + B_x$ $R_y = A_y + B_y$

1. 2 pts

Restate 310° in two other ways.

2. 2 pts

Restate 25° south of west in two other ways.

3. 2 pts

Restate −200° in two other ways.

4. 5 pts

Rey's speeder moves east at 52 m/s while a crosswind pushes it south at 36 m/s. Find the speeder's resultant velocity: magnitude and direction.

5. 5 pts

Dumbo flaps at 24 m/s toward 25° north of west while a gust blowing toward 40° east of north adds 9.0 m/s. Find Dumbo's resultant velocity: magnitude and direction.